

Real Time Poisoning Attacks and Privacy Strategies on Machine Learning Systems

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Abstract—Machine Learning aims to learn computer systems and predict output. Nowadays, Deep learning uses continuous training based on a large network of interconnected neurons to mimic the way humans think, analyze, and make decisions. It plays an important role in real-time applications. There are some security and privacy threats faced by deep learning systems. One of the major threats faced by such learning systems is attacks during training. The manipulation of training data leads to poisoning attacks. The resulting models are severely impacted by poisoning attacks, which may prevent them from converging or alter the outcomes of their predictions. The need to defend against such poisoning attacks is urgent and difficult. This study explains a thorough and detailed analysis of different types of poisoning attacks in real time applications. Its countermeasures to mitigate such attacks and various privacy-enhancing methods are also discussed here.

Index Terms—Federated Learning, Deep Learning, Convolutional Neural Networks, Generative Adversarial Networks (GAN)

I. INTRODUCTION

Machine learning (ML) is a research field that focuses on the theory, properties, and performance of learning, algorithms, and systems. Deep Learning is a subset of ML and AI technology which can be trained for performing classification and identify patterns in different types of data like text, image, voice etc. DL is used in fields such as Computer vision, Recommender systems, Healthcare systems, self-driving cars, etc. Google has integrated a deep learning-based vehicle detection technology. Its integrated laser detecting equipment can quickly determine the best route and the buffer distance while properly identifying obstacles including pedestrians, trains, and automobiles. The presence of adversaries affect the accuracy of ML models. As DL frameworks are progressively coordinated into security and privacy-preserving applications, adversarial attacks pose a significant risk.

Machine learning is used as a service in different companies to facilitate automated deep learning. With MLaaS, users may access a variety of training and prediction services by just uploading data from the local servers. But, outsourcing this

type of training raises several security and privacy issues. Those malicious attacks on the data cause misclassification in a machine-learning pipeline that is not visible to the human eye. Such attacks are called adversarial attacks which are affected in different data modalities such as text, image, audio, etc.

The data, model, training, testing, and validation processes are the main ML concerns for an AI system. Although AI also incorporates several knowledge-based strategies (Inference Systems), the statistical-driven strategy of ML poses unique security issues throughout the training and inference phases. The ML techniques work under the assumption that their surroundings are benign. However, this assumption is not always true. The possibility of manipulating training data and taking advantage of model sensitivity to impair ML classification and regression performance is one of these security problems [9].

Poisoning attacks occur in different scenarios which affect users' safety. In the case of self-driving car [37], the attacker may inject particular labels to the stop sign during the training phase. These samples would be treated as signs of speed limit by the ML model trained on the poisoned sign with an attacker-defined trigger, which causes the vehicle to continue to move. During the training phase, the laser pulse is spoofed by the attacker. In the testing phase, the deployed LiDAR sensors might identify a fake obstacle distance, making the situation unsafe for the passengers to drive in. This scenario is shown in Fig 1. [37]. Cameras used in autonomous vehicles are used for detecting and recognizing object to gather information. This data is passed through some training algorithms. RADAR [45] is a sensor which uses radio waves for computing distance measures. LiDAR is another type of sensor having light as a medium. Both these LiDAR and RADAR sensors helps to obtain important information to perceive the surroundings.

Adversarial examples such as data poisoning, online system manipulation, transfer learning attacks, overfitting, etc. might put ML models at risk. There are two main stages of ML systems: the training phase and the testing phase. In the case of content poisoning attacks [42], manipulated content is stored in the network cache and affects the default behavior of the

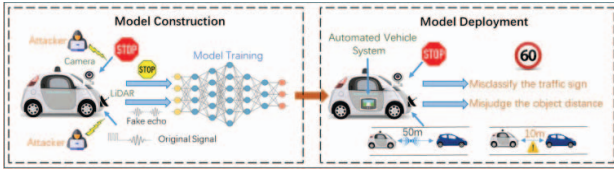


Fig. 1. Attacks in Self driving car[37]

system. In Named Data Networking systems, attackers add some fabricated data to the network node. Hence, the network becomes too congested and affects the transmission time from sender to receiver. Here, we discussed the detection methods of attacks during the training phase which affect the entire ML systems.

The model is trained, and the learning algorithm learns from the training data during the training phase. During the testing phase, the test data are given to a trained model to determine whether it is performing as expected. Attackers may attack during either of these crucial phases to obtain private information. The pipeline of a machine learning system contains numerous stages and regions where it is vulnerable. The manipulation of training data and model theft are both threats which affect the output of the model. It is clear from these examples that the ML model is ineffective in many areas.

In real-world applications, low-quality data and insufficient samples of data affect the training of ML models. One of the solutions is to distribute the data globally and train such global data. Such sensitive training data need more and more protection and privacy. Due to such issues, different countries established different laws such as the Health Insurance Portability and Accountability Act, Data Protection Regulation, etc. There are some disadvantages to enacting these laws such as data interruption and information silos.

To address this issue, a distributed ML framework called the Federated Learning System was developed by Google in 2016. Here, clients trained the global model using the local data. This model is updated to a particular server. The global model is updated by the server through averaging of all the updated models. Since local data is accessed only by owners, data protection is effective in FL systems. But still, there are different challenges in privacy and security.

Here, we discussed the need for detecting poisoning attacks in real-time ML systems, its open challenges, proposed methods, etc. The detailed description of various poisoning attacks and threat models are covered in Sections II and III. Section IV proposed different methods to mitigate such attacks. Some privacy-enhancing cryptographic methods, open challenges, proposed methods are also discussed in the rest of the paper.

II. RELATED WORKS

Nowadays, several works have been done related to poisoning attacks and their defensive mechanisms. Some recent developments related to such works are mentioned below in detail.

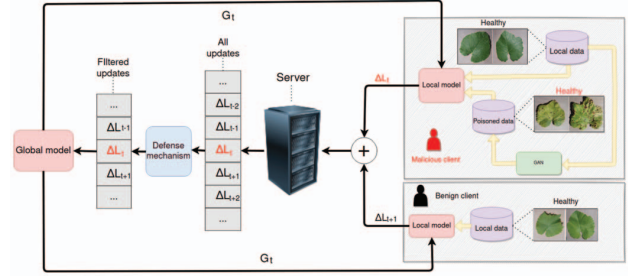


Fig. 2. GAN driven poisoning attack[2]

Akhtar et al.[4] discussed the review of various advanced developments in adversarial attacks and defense methods in vision applications. Different types of adversarial attacks on real-time applications such as real-world attacks, advanced gradient-based attacks, white box attacks, black box attacks, etc. are described. Randomized smoothing is one of the defense methods used. This technique injects noise into the classifier during the testing phase and is applied to some arbitrary functions. Hence, robustness can be ensured. Some other defense methods such as ensemble generative cleaning with a feedback loop, regularization scheme for ReLU networks and the use of Variational Auto-Encoders (VAEs) are also mentioned. Finding attacks in vision as well as language models is emerging as a tedious task.

Psychogyios et al.[2] discussed generating images using GAN. Here, label-flipping attacks are generated and tested based on an aggregation algorithm. Secure aggregation methods in FL help to improve the performance of ML applications. The entire process is illustrated in Fig. 2. The plantVillage dataset is used to test the data. By using defense methods such as Median aggregation, Trimmed mean aggregation, and Krum aggregation, those datasets are tested. Accuracy issues still pose a major challenge in this area. Adding additional datasets and hyperparameters also helps to improve accuracy.

The different types of data poisoning attacks which are discussed in various papers are described as follows:

A. Label Flipping Attacks[28]

It is a simple and effective way to flip labels of some training instances. This is either model-dependent or independent. Here, the attackers have no basic knowledge about the target ML model. These flipping attacks may be targeted or untargeted. It is based on model dependent. In a targeted attack, the attacker classifies wrongly one class consistently which results in the prediction of wrong outputs. Here, there is a large difference between the training instance and the margin of an SVM when it is flipped. In the case of untargeted attacks, the attacker chooses instances from some classes to misclassify as another class.

B. Clean Label Data Poisoning Attack[29]

The introduction of clean label data poisoning attacks compromise the security of DL networks. This is accomplished

TABLE I
SUMMARIZING EXISTING RECENT SURVEYS

Literature	Methods used	Challenges
[34]	Byzantine robust aggregation in Federated Learning Systems	Stealthiness and persistence of poisoning attacks
[13]	Ensemble-based approach against untargeted data poisoning	Perturbations added during label flipping method is not effective which causes performance degradation
[26]	Real-Time Poisoning attacks detection using Model Verification in deep computer vision	Method works on neural networks only
[41]	MobileNet model with Adam optimizer method for feature extraction,adaptive Neuro Fuzzy system for image classification with honey badger optimization algorithm	Performance Improvement by hybrid DeepLearning Models

by injecting a training set with samples of correctly labeled poison. The attacker manipulates only textual data instead of labelled data. Although user might perceive poison samples as being normal, they include irrelevant characteristics that can cause targeted misclassification throughout the inference process.

C. Feature Collision Attack[31]

Here, the attacker is not able to access training data and has no control over the labeling process. The attacker is unable to alter the target instance during inference and is unaware of the architecture and settings of the feature extractor.

D. Convex Polytope Attack and Bullseye Polytope Attack[30]

It is done by producing poison instances in a feature space that are identical to a specified target instance. In a black-box context, the bullseye polytope attack outperforms a feature collision attack in terms of attack success rate, and it improves a convex polytope attack in terms of speed and attack success rate during end-to-end training.

E. Backdoor Attack[33]

A type of poisoning attack that contains a backdoor trigger. Here, there are three attack scenarios. They are outsourced training, transfer learning, and Federated learning.

Geming Xia et al.[34] discussed the comprehensive survey of different poisoning attack scenarios in Federated Learning. The different types of poisoning attacks and defense strategies in FL systems are analyzed in detail. The process of FL method is described using the FedAVG algorithm. The defense strategies such as analysis based on model, Byzantine robust aggregation, and Verification methods are also discussed here. These three methods are not so efficient in the case of non-IID settings. When the individual client trains the local models which are different, it is difficult for the server to identify malicious updates from harmful ones. In the case of real-world

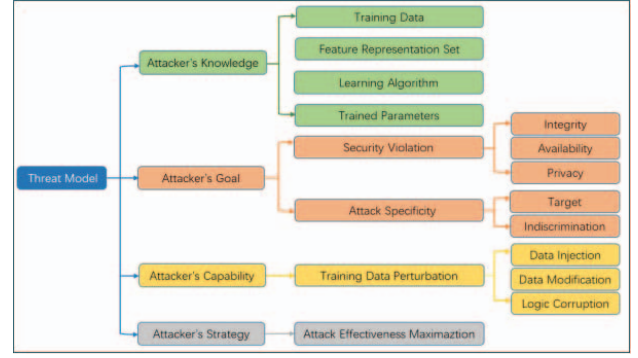


Fig. 3. Different threat models[37]

Attack	Attacker Capability	Attacker Goal	ML Models	Data Modality
Poisoning availability	Poison a large percentage of training data	Modify ML model indiscriminately	• Linear regression • Logistic regression, SVM, and DNNs	• Vision • Tabular data • Security
Backdoor poisoning	Insert a backdoor in training and testing data	Misclassify backdoored examples	• DNNs • LightGBM, DNNs, RF, and SVM	• Vision • Tabular data • Security
Targeted poisoning	Insert poisoned points in training	Misclassify targeted point	• DNNs • Word embeddings	• Vision • Text
Subpopulation poisoning	Identify subpopulation Insert poisoned points from subpopulation	Misclassify natural points from subpopulation	• Logistic regression and DNNs	• Vision • Tabular data • Text

DNN: deep neural network; RF: random forest.

Fig. 4. Different poisoning attacks[32]

scenarios, it is necessary to make sure whether the third party is trusted or not. So, the Verification approach needs further improvement. Table 1 shows the summary of some existing recent surveys.

III. THREAT MODEL

The discovery and evaluation of potential security risks and weaknesses in intelligent networking systems are ensured by threat modeling. Poisoning attacks concentrate on the changing of the information gathered and kept by objects or nodes in the system's perception layer. The knowledge of the attacker, their intent, how they plan to use the ML model, and their ability to affect the training data are all factors which is shown in Fig 3.

IV. POISONING ATTACKS AND THEIR COUNTERMEASURES

In order to modify the learned ML model and alter the predictions produced, attackers can purposely inject harmful samples during the training phase. To poison a model during training, poisoning attacks require altering training data. In recent surveys, the Federated Learning method has been used to protect the privacy of users. But, still FL also has security and privacy issues. Some of the countermeasures are listed in detail. Fig.4 lists four types of poisoning attacks along with the capabilities and objectives of the attacker, the ML models against which the attacks have been assessed, and the data modalities (such as visual, text, cyber security, and tabular data) to which they have been applied.

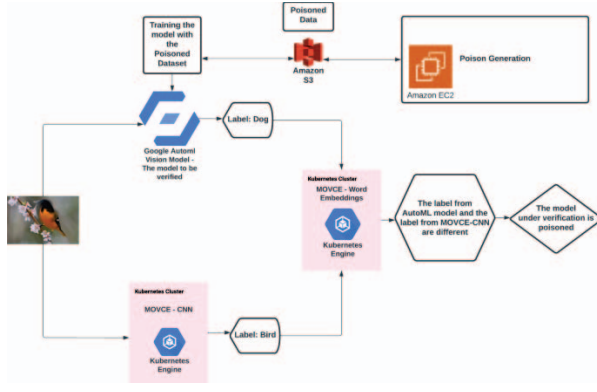


Fig. 5. Training of data using Model Verification with CNN[26]

A. Verification method [26]

This method is used to identify any threat before performing in real-time applications. Both CNN and word embedding layers are attached for detecting real-time poison attacks. Here, images are generated through autoML. The Fully Connected layer is used for classification and prediction. There are pooling layers followed by two fully connected layers. The model is then verified with word embedding and detected when poisoning occurs which is shown Fig 5.

B. GAN based poisoning[2]

In this method, images are trained using GAN. By using the dataset of the malicious clients, the adversary trains GAN and utilizes it to carry out label-flipping attacks. Detection can be made possible by calculating the distance between model parameters. To measure the dissimilarity measures, Euclidean distance, cosine distance, etc. are used. Some aggregation functions such as Median aggregation, Krum aggregation, and Trimmed mean aggregation methods are also used.

C. Detection using Data Washing and IDA algorithm[27]

The denoising autoencoder is used to initially send the data. Then, a little amount of Gaussian noise is added to data and the data are once more put through the autoencoder to produce the recovered data. The technique successfully removes the malicious signal produced by the attacker using the denoising autoencoder. The denoising autoencoder can remove some non-Gaussian noise in addition to its primary goal of converting a signal containing Gaussian noise into a clean signal. Detection of paralysis attacks and other types of attacks is done through the IDA algorithm. The process starts by looking for paralysis attacks. If no such attack is found, the system then checks to see whether any other types of attacks are present. The algorithm does a final check for a targeted attack in the absence of any obvious attacks being found. The process ends if the data continues to be normal. The steps are shown in Fig 6.

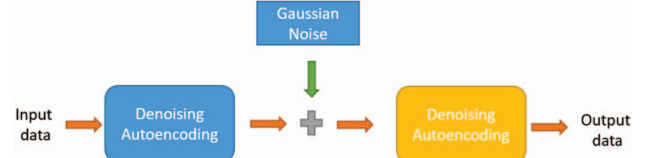


Fig. 6. Steps of Data washing algorithm [27]

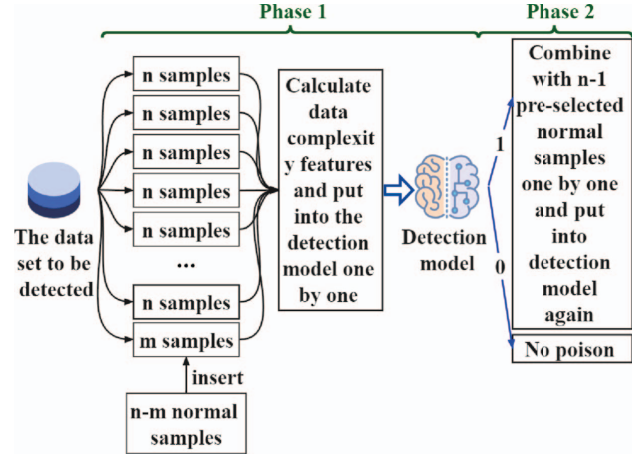


Fig. 7. Batch Detection Method[44]

D. Sanitization of Training data[44]

One of the defense methods against poisoning attacks is the sanitization of training data. This method is suitable in the FL environment since data sanitization is done before local model training. Here, the batch detection method is used in the data sanitization process. There are two different phases for the batch detection process which is shown in Fig 7. In Phase 1, data complexity features are calculated and applied to the model batch-wise. In phase 2, each sample is combined with another $n-1$ normal sample to generate n subsets. Poisoned samples are detected during phase 2. These sanitized models are trained by FL clients independently. This method helps to detect poison before training locally. Defense against feature perturbation poisons is still a challenging factor.

V. PRIVACY ENHANCING METHODS

A. Chained Secure Multi-party Computing

In chained secure MPC[22], individuals can send masked information using a serial chain frame. The architecture consists of a server that is responsible for sending messages simultaneously to the group of users before training. Random numbers are generated by the server which indicates the value of the mask. Generate tokens, summation of the results and average, then aggregate results and tokens are extracted and finally distribute the global model parameters are sent back to users which is shown in Fig.8

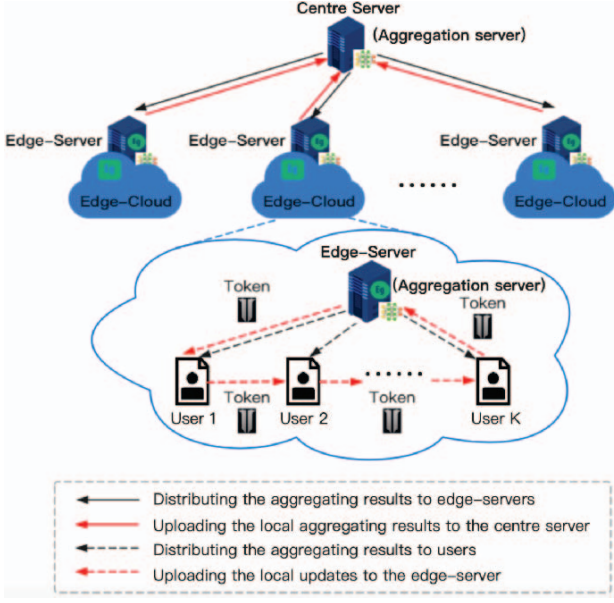


Fig. 8. Architecture of Chain-PPFL[22]

B. Differential Privacy

Differential privacy[24] is used to protect users' privacy in datasets. By facilitating data analysis without revealing sensitive information about any of the individuals in the dataset, it may provide an accurate guarantee of privacy. DP offers a guarantee of privacy protection that can be mathematically proven against a variety of privacy attacks.

Samples of sensitive training data are given as input to the model, and gradients are computed using the Stochastic Gradient Descent Method (SGD). Additionally, noise is added to this gradient model before updating the model. Differential privacy protects an individual's privacy by adding some random noise into the dataset. This causes the results to become different at different times. The privacy loss parameter determines the amount of noise. It can be computed from the Laplace distribution. Fig. 9 illustrates the differential privacy method used in the FL system. Here, the Server starts by initializing the FL model and sends parameters to different clients. Then each client undergoes training using the parameters sent from the server. Next, the model clipping procedure is done and generates clipped model parameters. In the next step, each client performs model perturbation, and the perturbed local models are sent to the server. In the end, using the aggregation rule server generates the FL aggregate model parameter and sent to each client.

C. Homomorphic Encryption

An encryption method known as homomorphic encryption (HE) enables computation on encrypted data as if it were unprotected plain text. According to the mathematical principle of homomorphism, a computation's structure remains unchanged. This method is mostly applied in CNN. It can also

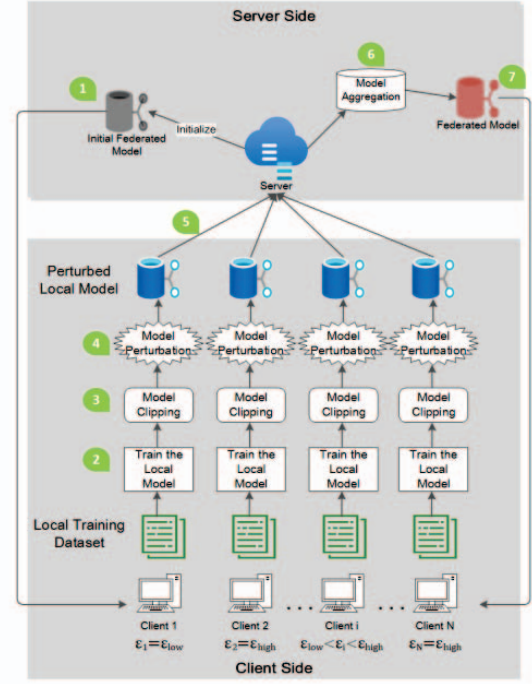


Fig. 9. Differential Privacy[43]

be used to securely aggregate encrypted algorithm updates in federated learning environments. Here, the plain text is encrypted using either RSA or ElGamal scheme. The generated cipher text is sent to the distributed learning environment. This encrypted data is again encrypted and the result is stored in the cloud. There are two types of homomorphic encryptions. If the multiplicative/additive operations are applied to the encrypted data, then it is partially homomorphic encryption. If both operations are used on the encrypted data, then it is fully homomorphic encryption.

D. Combination of PVD and RSA Algorithm

Image security during training can be done using the combination of Pixel value differencing steganography[40] and RSA algorithm. Here, multidimensional images can be secured using this scheme. Data hiding is done by sending secret bits to different directions of the images. Thus, security can be ensured.

VI. OPEN CHALLENGES

Although there are different defense strategies against poisoning attacks, there are still some open challenges. Robustness in optimization methods is one of the problems faced by ML systems. Data privacy is another major challenge faced by healthcare applications. Nowadays these decentralized systems are used in various applications including banking, and the protection of data is still an open challenging problem. Defense against backdoor poisoning is still an open problem in vision systems. Another challenge faced by the FL system

TABLE II
CHALLENGES IN DISTRIBUTED ML SYSTEMS

Challenges	Description
The balance between ML performance and security level	trade-off between learning performance, convergence, security, and privacy level should be well designed.
Decentralized Paradigm and communication among multiple participants should be investigated	Communication among multiple participants should be investigated
Complexity Reduction	Computational complexity and security among distributed systems still be the issue.

is Federated Poisoning Attacks. If a client pretends to be a benign participant, and poisons the local and global model, then it will affect the prediction of the whole ML system. So, such attacks remain an open problem. Some other challenges are shown in TABLE II:

VII. PROPOSED METHODS

The necessity of poisoning attack detection in real-time applications is very promising nowadays. An integrated framework of the Model Verification Method along with a transfer learning network can be used for the detection of attacks on real-time ML systems. Computational costs can be optimized using swarm optimization techniques. Its privacy can be enhanced using the hybrid method of RSA and homomorphic encryption techniques.

VIII. CONCLUSION

Machine learning applications are increasing nowadays. Here, we discussed the survey of various types of adversary attacks, defense measures, application areas, and privacy-enhancing methods in ML systems. Some proposed methods are also suggested for the efficient working of real-time ML systems. Real-time applications need more security and accuracy. So, the necessity of detecting and preventing such attacks is high. Although different defense strategies are proposed to detect such attacks, there are still some open challenges. So, it is necessary to build more sophisticated methods for enhancing the security and privacy of ML models and data.

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